

Creation Date 01-Sep-2009

Revision Date 02-May-2025

Revision Number 2

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                                  |                                                              |
|----------------------------------|--------------------------------------------------------------|
| <b>Product Description:</b>      | <b>Isopropanol, VLSI electronic grade</b>                    |
| <b>Cat No. :</b>                 | <b>S60414</b>                                                |
| <b>Synonyms</b>                  | 2-Propanol; IPA; Isopropyl alcohol; Propan-2-ol; Isopropanol |
| <b>Index No</b>                  | 603-117-00-0                                                 |
| <b>CAS No</b>                    | 67-63-0                                                      |
| <b>EC No</b>                     | 200-661-7                                                    |
| <b>Molecular Formula</b>         | C3 H8 O                                                      |
| <b>REACH registration number</b> | 01-2119457558-25-0196                                        |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                                       |                                                                                             |
|---------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Recommended Use</b>                | Laboratory chemicals.                                                                       |
| <b>Sector of use</b>                  | SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites    |
| <b>Product category</b>               | PC21 - Laboratory chemicals                                                                 |
| <b>Process categories</b>             | PROC15 - Use as a laboratory reagent                                                        |
| <b>Environmental release category</b> | ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates) |
| <b>Uses advised against</b>           | No Information available                                                                    |

### 1.3. Details of the supplier of the safety data sheet

|                |                                                                                                                                                                                                              |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Company</b> | Avocado Research Chemicals Ltd.<br>(Part of Thermo Fisher Scientific)<br>Shore Road, Heysham<br>Lancashire, LA3 2XY,<br>United Kingdom<br>Office Tel: +44 (0) 1524 850506<br>Office Fax: +44 (0) 1524 850608 |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                       |                                |
|-----------------------|--------------------------------|
| <b>E-mail address</b> | begel.sdsdesk@thermofisher.com |
|-----------------------|--------------------------------|

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

**GHS Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567**

**Physical hazards**

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|                                                                  |                   |
|------------------------------------------------------------------|-------------------|
| Flammable liquids                                                | Category 2 (H225) |
| <b>Health hazards</b>                                            |                   |
| Serious Eye Damage/Eye Irritation                                | Category 2 (H319) |
| Specific target organ toxicity - (single exposure)               | Category 3 (H336) |
| <b>Environmental hazards</b>                                     |                   |
| Based on available data, the classification criteria are not met |                   |

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

Danger

### Hazard Statements

- H225 - Highly flammable liquid and vapor
- H319 - Causes serious eye irritation
- H336 - May cause drowsiness or dizziness

### Precautionary Statements

- P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking
- P240 - Ground and bond container and receiving equipment
- P261 - Avoid breathing dust/fume/gas/mist/vapors/spray
- P280 - Wear protective gloves/protective clothing/eye protection/face protection
- P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

## 2.3. Other hazards

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

| Component         | CAS No  | EC No     | Weight % | GHS Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567 |
|-------------------|---------|-----------|----------|-----------------------------------------------------------------------------------------|
| Isopropyl alcohol | 67-63-0 | 200-661-7 | >95      | Flam. Liq. 2 (H225)<br>Eye Irrit. 2 (H319)<br>STOT SE 3 (H336)                          |

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Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                    |                                                                                                                                                  |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Eye Contact                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                  |
| Skin Contact                       | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention if symptoms occur.                                      |
| Ingestion                          | Do NOT induce vomiting. Get medical attention.                                                                                                   |
| Inhalation                         | Remove to fresh air. Get medical attention. If not breathing, give artificial respiration.                                                       |
| Self-Protection of the First Aider | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

### 4.2. Most important symptoms and effects, both acute and delayed

Difficulty in breathing. May cause central nervous system depression: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

### 4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically. Symptoms may be delayed.

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### Suitable Extinguishing Media

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

#### Extinguishing media which must not be used for safety reasons

Do not use water jetstream. Do not use a solid water stream as it may scatter and spread fire.

### 5.2. Special hazards arising from the substance or mixture

Flammable. Risk of ignition. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated.

#### Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), peroxides.

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

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## **6.1. Personal precautions, protective equipment and emergency procedures**

Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges. Avoid contact with skin, eyes or clothing.

## **6.2. Environmental precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information.

## **6.3. Methods and material for containment and cleaning up**

Prevent further leakage or spillage if safe to do so. Remove all sources of ignition. Soak up with inert absorbent material. Take precautionary measures against static discharges. Use spark-proof tools and explosion-proof equipment. Keep in suitable, closed containers for disposal.

## **6.4. Reference to other sections**

Refer to protective measures listed in Sections 8 and 13.

## **SECTION 7: HANDLING AND STORAGE**

### **7.1. Precautions for safe handling**

Wear personal protective equipment/face protection. Keep away from open flames, hot surfaces and sources of ignition. Use spark-proof tools and explosion-proof equipment. Use only non-sparking tools. Take precautionary measures against static discharges. Do not get in eyes, on skin, or on clothing. Do not breathe mist/vapors/spray. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded.

### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### **7.2. Conditions for safe storage, including any incompatibilities**

Keep away from heat, sparks and flame. Flammables area. Keep container tightly closed in a dry and well-ventilated place.

**Technical Rules for Hazardous Substances (TRGS) 510**      Class 3  
**Storage Class (LGK) (Germany)**

### **7.3. Specific end use(s)**

Use in laboratories

## **SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

### **8.1. Control parameters**

#### **Exposure limits**

List source(s): **UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020. **IRE** - 2021 Code of Practice for the Chemical Agents Regulations, Schedule 1. Published by the Health and Safety Authority

| Component         | The United Kingdom   | European Union | Ireland            |
|-------------------|----------------------|----------------|--------------------|
| Isopropyl alcohol | STEL: 500 ppm 15 min |                | TWA: 200 ppm 8 hr. |

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|  |                                                                                             |  |                              |
|--|---------------------------------------------------------------------------------------------|--|------------------------------|
|  | STEL: 1250 mg/m <sup>3</sup> 15 min<br>TWA: 400 ppm 8 hr<br>TWA: 999 mg/m <sup>3</sup> 8 hr |  | STEL: 400 ppm 15 min<br>Skin |
|--|---------------------------------------------------------------------------------------------|--|------------------------------|

## Biological limit values

List source(s):

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                            | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|--------------------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Isopropyl alcohol<br>67-63-0 ( >95 ) |                              |                                 |                                | DNEL = 888mg/kg<br>bw/day         |

| Component                            | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|--------------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Isopropyl alcohol<br>67-63-0 ( >95 ) |                                  |                                     |                                    | DNEL = 500mg/m <sup>3</sup>           |

## Predicted No Effect Concentration (PNEC)

According to our experience and to the information provided to us, the product does not have any harmful effects if it is used and handled as specified. See values below.

| Component                            | Fresh water      | Fresh water sediment           | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)        |
|--------------------------------------|------------------|--------------------------------|--------------------|------------------------------------|---------------------------|
| Isopropyl alcohol<br>67-63-0 ( >95 ) | PNEC = 140.9mg/L | PNEC = 552mg/kg<br>sediment dw | PNEC = 140.9mg/L   | PNEC = 2251mg/L                    | PNEC = 28mg/kg<br>soil dw |

| Component                            | Marine water     | Marine water sediment          | Marine water intermittent | Food chain              | Air |
|--------------------------------------|------------------|--------------------------------|---------------------------|-------------------------|-----|
| Isopropyl alcohol<br>67-63-0 ( >95 ) | PNEC = 140.9mg/L | PNEC = 552mg/kg<br>sediment dw |                           | PNEC = 160mg/kg<br>food |     |

## 8.2. Exposure controls

### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time   | Glove thickness | EU standard | Glove comments                                                                 |
|----------------|---------------------|-----------------|-------------|--------------------------------------------------------------------------------|
| Butyl rubber   | > 480 minutes       | 0.5 mm          | EN 374      | Permeation rate < 0.9 µg/cm <sup>2</sup> /min                                  |
| Nitrile rubber | > 360 - 480 minutes | 0.35 - 0.55 mm  |             | As tested under EN374-3 Determination of Resistance to Permeation by Chemicals |
| Viton (R)      | > 480 minutes       | 0.4 mm          |             |                                                                                |
| Neoprene       | < 40 minutes        | 0.7 mm          |             |                                                                                |

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

Inspect gloves before use.

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Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

## Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

## Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

## Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Valve filtering: EN405; Half mask: EN140; plus filter, EN 141  
When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls** No information available.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                                |                                             |                                                                                       |
|------------------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Physical State</b>                          | Liquid                                      |                                                                                       |
| <b>Appearance</b>                              | Colorless                                   |                                                                                       |
| <b>Odor</b>                                    | Alcohol-like                                |                                                                                       |
| <b>Odor Threshold</b>                          | No data available                           |                                                                                       |
| <b>Melting Point/Range</b>                     | -89.5 °C / -129.1 °F                        |                                                                                       |
| <b>Softening Point</b>                         | No data available                           |                                                                                       |
| <b>Boiling Point/Range</b>                     | 81 - 83 °C / 177.8 - 181.4 °F               | @ 760 mmHg                                                                            |
| <b>Flammability (liquid)</b>                   | Highly flammable                            | On basis of test data                                                                 |
| <b>Flammability (solid,gas)</b>                | Not applicable                              | Liquid                                                                                |
| <b>Explosion Limits</b>                        | <b>Lower</b> 2 Vol%<br><b>Upper</b> 12 Vol% |                                                                                       |
| <b>Flash Point</b>                             | 12 °C / 53.6 °F                             | <b>Method -</b> Abel Closed Cup (BS 2000 Part 170, IP 170, AS/NZS 2106)<br>ASTM E-659 |
| <b>Autoignition Temperature</b>                | 425 °C / 797 °F                             |                                                                                       |
| <b>Decomposition Temperature</b>               | No data available                           |                                                                                       |
| <b>pH</b>                                      | 7                                           | 1% aq. sol                                                                            |
| <b>Viscosity</b>                               | 2.27 mPa.s at 20 °C                         |                                                                                       |
| <b>Water Solubility</b>                        | Miscible                                    |                                                                                       |
| <b>Solubility in other solvents</b>            | No information available                    |                                                                                       |
| <b>Partition Coefficient (n-octanol/water)</b> |                                             |                                                                                       |
| <b>Component</b>                               | <b>log Pow</b>                              |                                                                                       |
| Isopropyl alcohol                              | 0.05                                        |                                                                                       |
| <b>Vapor Pressure</b>                          | 43 mmHg @ 20 °C                             |                                                                                       |
| <b>Density / Specific Gravity</b>              | 0.785                                       | ASTM D-4052                                                                           |
| <b>Bulk Density</b>                            | Not applicable                              | Liquid                                                                                |
| <b>Vapor Density</b>                           | 2.1 @ 20 °C / 68 °F                         | (Air = 1.0)                                                                           |
| <b>Particle characteristics</b>                | Not applicable (liquid)                     |                                                                                       |

### 9.2. Other information

**Molecular Formula** C3 H8 O

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|                                 |                                                                                                  |
|---------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Molecular Weight</b>         | 60.1                                                                                             |
| <b>VOC Content(%)</b>           | 100% (Organic Carbon (by mass) = 59.9 %) (EC/1999/13)                                            |
| <b>Explosive Properties</b>     | Not explosive explosive air/vapour mixtures possible Vapors may form explosive mixtures with air |
| <b>Evaporation Rate</b>         | 1.7 - ASTM D 3539 (Butyl acetate = 1.0)                                                          |
| <b>Thermal conductivity</b>     | 0.137 W/m °C at 20 °C / 68 °F                                                                    |
| <b>Refractive index</b>         | 1.377 at 20 °C / 68 °F (ASTM D-1218)                                                             |
| <b>Surface tension</b>          | 22.7 mN/m at 20 °C / 68 °F                                                                       |
| <b>Coefficient of expansion</b> | 0.0009 / °C                                                                                      |
| <b>Specific heat capacity</b>   | 3 kJ/kg °C at 20 °C / 68 °F                                                                      |
| <b>Dielectric constant</b>      | 18.6 at 20 °C / 68 °F                                                                            |
| <b>Heat of vapourisation</b>    | 665 J/g                                                                                          |

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

#### **Hazardous Polymerization**

Hazardous polymerization does not occur.

#### **Hazardous Reactions**

None under normal processing.

### 10.4. Conditions to avoid

Heat, flames and sparks. Keep away from open flames, hot surfaces and sources of ignition.

### 10.5. Incompatible materials

Strong oxidizing agents. Acids. Halogens. Acid anhydrides.

### 10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). peroxides.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### **Product Information**

#### **(a) acute toxicity;**

##### **Oral**

Based on available data, the classification criteria are not met

##### **Dermal**

Based on available data, the classification criteria are not met

##### **Inhalation**

Based on available data, the classification criteria are not met

| Component         | LD50 Oral                                  | LD50 Dermal         | LC50 Inhalation       |
|-------------------|--------------------------------------------|---------------------|-----------------------|
| Isopropyl alcohol | 5045 mg/kg ( Rat )<br>3600 mg/kg ( Mouse ) | 12800 mg/kg ( Rat ) | 72.6 mg/L ( Rat ) 4 h |

#### **(b) skin corrosion/irritation;**

Based on available data, the classification criteria are not met

#### **(c) serious eye damage/irritation;**

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**(d) respiratory or skin sensitization;**

**Respiratory**

Based on available data, the classification criteria are not met

**Skin**

Based on available data, the classification criteria are not met

**(e) germ cell mutagenicity;**

Based on available data, the classification criteria are not met

**(f) carcinogenicity;**

Based on available data, the classification criteria are not met

There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;**

Based on available data, the classification criteria are not met

**(h) STOT-single exposure;**

Category 3

**Results / Target organs**

Central nervous system (CNS).

**(i) STOT-repeated exposure;**

Based on available data, the classification criteria are not met

**Target Organs**

None known.

**(j) aspiration hazard;**

Based on available data, the classification criteria are not met

**Symptoms / effects, both acute and delayed**

May cause central nervous system depression. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

**11.2. Information on other hazards**

**Endocrine Disrupting Properties**

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

**12.1. Toxicity**

**Ecotoxicity effects**

. Do not empty into drains.

| Component         | Freshwater Fish                                                                                                                                                                                                          | Water Flea                                      | Freshwater Algae                                                                                           |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Isopropyl alcohol | LC50: = 9640 mg/L, 96h<br>flow-through (Pimephales promelas)<br>LC50: > 1400000 µg/L, 96h<br>(Lepomis macrochirus)<br>LC50: = 11130 mg/L, 96h static<br>(Pimephales promelas)<br>LC50: = 10000000 µg/L, 96h<br>(Daphnia) | 13299 mg/L EC50 = 48 h<br>9714 mg/L EC50 = 24 h | EC50: > 1000 mg/L, 72h<br>(Desmodesmus subspicatus)<br>EC50: > 1000 mg/L, 96h<br>(Desmodesmus subspicatus) |

| Component         | Microtox                                              | M-Factor |
|-------------------|-------------------------------------------------------|----------|
| Isopropyl alcohol | = 35390 mg/L EC50 Photobacterium phosphoreum<br>5 min |          |

**12.2. Persistence and degradability**

**Persistence**

Expected to be biodegradable

Persistence is unlikely, based on information available.



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## 12.3. Bioaccumulative potential

Bioaccumulation is unlikely

| Component         | log Pow | Bioconcentration factor (BCF) |
|-------------------|---------|-------------------------------|
| Isopropyl alcohol | 0.05    | No data available             |

## 12.4. Mobility in soil

The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air.

### Surface tension

22.7 mN/m at 20 °C / 68 °F

## 12.5. Results of PBT and vPvB assessment

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).

## 12.6. Endocrine disrupting properties

### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

## 12.7. Other adverse effects Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance

This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

#### Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

#### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

#### European Waste Catalogue (EWC)

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

#### Other Information

Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations.

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

#### 14.1. UN number

UN1219

#### 14.2. UN proper shipping name

Isopropanol (Isopropyl alcohol)

#### 14.3. Transport hazard class(es)

3

#### 14.4. Packing group

II

### ADR

#### 14.1. UN number

UN1219

#### 14.2. UN proper shipping name

Isopropanol (Isopropyl alcohol)

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**14.3. Transport hazard class(es)** 3  
**14.4. Packing group** II

## IATA

**14.1. UN number** UN1219  
**14.2. UN proper shipping name** Isopropanol  
**14.3. Transport hazard class(es)** 3  
**14.4. Packing group** II

**14.5. Environmental hazards** No hazards identified

**14.6. Special precautions for user** No special precautions required.

**14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component         | CAS No  | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|-------------------|---------|-----------|--------|-----|-------|------|----------|------|------|
| Isopropyl alcohol | 67-63-0 | 200-661-7 | -      | -   | X     | X    | KE-29363 | X    | X    |

| Component         | CAS No  | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|-------------------|---------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Isopropyl alcohol | 67-63-0 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### Authorisation/Restrictions according to EU REACH

| Component         | CAS No  | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-------------------|---------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Isopropyl alcohol | 67-63-0 | -                                                                   | Use restricted. See entry 75. (see link for restriction details)              | -                                                                                                     |

#### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

### Seveso III Directive (2012/18/EC)

| Component         | CAS No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-------------------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Isopropyl alcohol | 67-63-0 | Not applicable                                                                            | Not applicable                                                                           |

**Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals**  
Not applicable

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## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

See table for values

| Component         | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|-------------------|---------------------------------------|-------------------------|
| Isopropyl alcohol | WGK1                                  |                         |

| Component         | France - INRS (Tables of occupational diseases)      |
|-------------------|------------------------------------------------------|
| Isopropyl alcohol | Tableaux des maladies professionnelles (TMP) - RG 84 |

| Component                            | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Isopropyl alcohol<br>67-63-0 ( >95 ) |                                                                                                                | Group I                                                                         |                                                                                             |

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has been conducted by the manufacturer/importer

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H225 - Highly flammable liquid and vapor

H319 - Causes serious eye irritation

H336 - May cause drowsiness or dizziness

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

# SAFETY DATA SHEET

Isopropanol, VLSI electronic grade

Revision Date 02-May-2025

**NOEC** - No Observed Effect Concentration  
**PBT** - Persistent, Bioaccumulative, Toxic

**POW** - Partition coefficient Octanol:Water  
**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

**Prepared By**

Health, Safety and Environmental Department

**Creation Date**

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**Revision Summary**

Initial Release.

**This safety data sheet complies with Regulation UK SI 2019/758 and UK SI 2020/1577 as amended.**

**Disclaimer**

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**End of Safety Data Sheet**